Exponential contraction and quadratic remainder for the linearized nonlinear recombination evolution

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Editorial note. Only Theorem wasscontrlemma2 is counted. The retained nonlinear contraction theorem and Wild-tree calculations are uncounted author support.

Allowed standard prerequisites (editorial).

- Finite-dimensional probability ODEs and uniqueness. Interpret the recombination semigroup; the Wild expansion and tree representation are proved local ingredients, not supplied conclusions.
- Total-variation optimal coupling and dual characterization. TV distance equals minimum mismatch probability and one half of the sup over bounded test functions of norm<=1.. Couple independent leaf configurations and convert their mismatch estimate to TV.
- Nonnegative series and elementary integral/ODE manipulation. Sum the finite-tree renewal identity and solve its resulting scalar linear ODE.

0.1 The nonlinear equation

Let $\Omega = \prod_{i=1}^n X_i$ be the set of n-vectors $\sigma = (\sigma_1, \ldots, \sigma_n)$ where $\sigma_i \in X_i$, and the X_i are given finite sets. We interpret σ as a particle. Thus, a particle is a string of n characters each taken from a finite alphabet. A basic example is obtained by taking $\Omega = \{0,1\}^n$. Without loss of generality we will assume that each X_i has the form $X_i := \{0, 1, 2, \ldots, q_i\}$, for some $q_i \in \mathbb{N}$. Given a subset $A \subset [n]$, $[n] = \{1, \ldots, n\}$, and $\sigma \in \Omega$, σ_A denotes the $A-$component of σ, that is the string $(\sigma_i, i \in A)$. If $(\sigma, \eta) \in \Omega \times \Omega$ is a pair of particles, the recombination at A consists in exchanging the A-component of σ with the A-component of η. This defines the map

$$(\sigma, \eta) \mapsto (\eta_A \sigma_{A^c}, \sigma_A \eta_{A^c}),$$

where $\eta_A \sigma_{A^c}$ denotes the element of Ω with entries η_i for $i \in A$ and σ_i for $i \in A^c = [n] \setminus A$. Let $\mathcal{P}(\Omega)$ denote the set of probability measures on Ω. If the original pair (σ, η) is obtained by sampling independently from $p \in \mathcal{P}(\Omega)$, then the new particle $\eta_A \sigma_{A^c}$ is distributed according to $p_A \otimes p_{A^c}$, the product of the

two marginals of p on the A and A^c components respectively. By choosing the set $A \subset [n]$ according to some distribution ν, one obtains the quadratic collision kernel

$$p \mapsto Q(p) = \sum_{A \subset [n]} \nu(A)\, p_A \otimes p_{A^c}.$$

The nonlinear evolution is defined by the dynamical system $\dot{p}_t = Q(p_t) - p_t$, that is

$$\frac{d}{dt}\, p_t = \sum_{A \subset [n]} \nu(A)\, (p_{t,A} \otimes p_{t,A^c} - p_t) \tag{1}$$

with the initial condition $p_0 = p \in \mathcal{P}(\Omega)$. Here $p_t \in \mathcal{P}(\Omega)$ is the distribution of the particle at time t, $p_{t,A}$ denotes its marginal on A and ν is a given probability measure over the subsets of $[n]$. The study of this model starts with the pioneering work of Geiringer [21]; see also [29, 39, 38, 40] for more recent accounts. It is well known that the Cauchy problem associated to (1) has a unique solution for every initial distribution $p \in \mathcal{P}(\Omega)$; see e.g. [40]. Moreover, it is not difficult to see that the evolution preserves the single site marginals, that is $p_{t,i} = p_i$ for all $t \geqslant 0$ and for all $i \in [n]$. We say that the recombination measure ν is *separating* if for any $i, j \in [n]$ there is a positive probability that the random set A with distribution ν separates i and j. Equivalently, if $r(\nu) < 1$ where

$$r(\nu) := \max_{i < j \in [n]} \nu\left(\{i,j\} \subset A \text{ or } \{i,j\} \subset A^c\right) \tag{2}$$

$$W(p,q) := \inf_{\Gamma \in \Pi(p,q)} \sum_{\sigma,\, \sigma' \in \Omega} \Gamma(\sigma, \sigma') \sum_{i=1}^{n} \mathbf{1}_{\sigma_i \neq \sigma'_i}, \tag{3}$$

where $p, q \in \mathcal{P}(\Omega)$, and $\Pi(p,q)$ denotes the set of all couplings Γ of p and q. We remark that the total variation distance $\|p - q\|_{\mathrm{TV}}$ is defined as above with $\sum_{i=1}^{n} \mathbf{1}_{\sigma_i \neq \sigma'_i}$ replaced by $\mathbf{1}_{\sigma \neq \sigma'}$ and thus one has

$$\|p - q\|_{\mathrm{TV}} \leqslant W(p,q) \leqslant n\, \|p - q\|_{\mathrm{TV}}. \tag{4}$$

Consider the symmetric bilinear form $\widehat{Q}(f,g)$ defined by

$$\begin{aligned} &\widehat{Q}(f,g)(\eta) \\ &= \frac{1}{2} \sum_{A,\sigma} \nu(A) \big(f(\sigma_A \eta_{A^c}) g(\eta_A \sigma_{A^c}) + g(\sigma_A \eta_{A^c}) f(\eta_A \sigma_{A^c}) - f(\sigma) g(\eta) - g(\sigma) f(\eta) \big), \end{aligned} \tag{5}$$

where $\eta \in \Omega$, $f, g : \Omega \mapsto \mathbb{R}$, and the sum extends to all $A \subset [n]$ and $\sigma \in \Omega$. If $p \in \mathcal{P}(\Omega)$, then $\widehat{Q}(p,p) = Q(p)$, where

$$Q(p) = \sum_{A} \nu(A)(p_A \otimes p_{A^c} - p)$$

is the generator associated to (1), that is the non linear semigroup $\{S_t, t \geqslant 0\}$ satisfies $S_0(p) = p$ and $\partial_t S_t(p) = Q(S_t(p))$, $t \geqslant 0$. Next, consider the differential equation

$$\partial_t h_t = 2\widehat{Q}(q_t, h_t), \qquad h_0 = h, \tag{6}$$

where $q_t = S_t(q)$ for some fixed $q \in \mathcal{P}(\Omega)$. Note that it is linear in h. We write its unique solution as

$$h_t = \bar{S}_t(q)(h),$$

that is $\bar{S}_t(q)(h)$ verifies $\partial_t \bar{S}_t(q)(h) = 2\widehat{Q}(q_t, \bar{S}_t(q)(h))$, $\bar{S}_0(q)(h) = h$ and

$$\bar{S}_t(q)(\bar{S}_s(q)(h)) = \bar{S}_s(q)(\bar{S}_t(q)(h)) = \bar{S}_{t+s}(q)(h).$$

Our main result concerning the linearized equation reads as follows.

Theorem 0.1. *For all $p, q \in \mathcal{P}(\Omega)$ such that $p_i = q_i$ for all $i \in [n]$, and $\phi : \Omega \mapsto \mathbb{R}$, for all $t > 0$,*

$$\left\langle \bar{S}_t(q)(p-q), \phi \right\rangle \leqslant n(n-1)\|\phi\|_\infty e^{-D(\nu)\, t}\, \|p-q\|_{\mathrm{TV}}, \tag{7}$$

$$\begin{aligned} &\left\langle S_t(p) - S_t(q) - \bar{S}_t(q)(p-q), \phi \right\rangle \\ &\qquad\qquad \leqslant \frac{1}{16}\, n^3(n-1)(n-2)\|\phi\|_\infty e^{-D(\nu)\, t}\, \|p-q\|_{\mathrm{TV}}^2, \end{aligned} \tag{8}$$

where $D(\nu) = 1 - r(\nu)$.

0.2 Wild sums and McKean trees

Given $f, g : \Omega \to \mathbb{R}$, we adopt the notation

$$(f \circ g)^A := \frac{1}{2}\left(f_A \otimes g_{A^c} + g_A \otimes f_{A^c}\right), \tag{9}$$

$$f \circ g := \sum_{A \subset [n]} \nu(A)(f \circ g)^A, \tag{10}$$

where $f_A(\sigma_A) := \sum_{\sigma_{A^c}} f(\sigma_A \sigma_{A^c})$. Then the nonlinear equation (1) is given by

$$\frac{d}{dt}\, p_t = p_t \circ p_t - p_t. \tag{11}$$

The convolution product defined by $f \circ g$ is commutative and distributive, but not associative. Following Wild's original construction [44] we write the solution of (11) with initial datum $p_0 = p$, as

$$p_t = e^{-t} \sum_{k=1}^{\infty} \left(1 - e^{-t}\right)^{k-1} p^{(k)}, \tag{12}$$

where the $\{p^{(k)}\}_{k \geqslant 1}$ are probability measures on Ω defined inductively by

$$p^{(1)} = p, \qquad p^{(k)} = \frac{1}{k-1} \sum_{j=1}^{k-1} p^{(j)} \circ p^{(k-j)}, \quad k \geqslant 2. \tag{13}$$

The validity of (12) can be easily checked by direct inspection, see e.g. the argument after (45) below for a similar computation.

Moreover, following McKean [31], we may express $p^{(k)}$ as a weighted sum over rooted binary trees γ with k leaves. We now recall the details of this construction, and refer the reader to [7] for further background. Let $\Gamma(k)$ denote the set of rooted binary trees with k leaves and call $\alpha_k(\gamma)$, $\gamma \in \Gamma(k)$, the probability over $\Gamma(k)$ obtained by the following procedure. $\Gamma(1) = \{\gamma_1\}$ is just the empty tree with only the root with $\alpha_1(\gamma_1) = 1$, and $\Gamma(2) = \{\gamma_2\}$ where γ_2 is the unique tree obtained by adding two children to the root, with $\alpha_2(\gamma_2) = 1$. Then, recursively, for $k \geqslant 2$, for any $\gamma_{k-1} \in \Gamma(k-1)$, consider all possible trees $\gamma_{k-1}^i \in \Gamma(k)$, $i = 1, \ldots, k-1$, obtained by adding two children to the ith leaf of γ_{k-1}, and for any $\gamma \in \Gamma(k)$, set

$$\alpha_k(\gamma) = \sum_{\gamma_{k-1} \in \Gamma(k-1)} \alpha_{k-1}(\gamma_{k-1}) \sum_{i=1}^{k-1} \frac{\mathbf{1}\left(\gamma = \gamma_{k-1}^i\right)}{k-1} \,.$$

This defines a probability α_k on $\Gamma(k)$, for any $k \in \mathbb{N}$, and one checks, recursively, that for all $k \geqslant 2$,

$$\alpha_k(\gamma) = \frac{1}{k-1} \alpha_j(\gamma_l) \alpha_{k-j}(\gamma_r), \tag{14}$$

where γ_l and γ_r denote respectively the subtree of γ rooted at the left child of the root and the subtree of γ rooted at the right child of the root, while j denotes the number of leaves in γ_l. Then, by induction over k it follows that for all $k \in \mathbb{N}$,

$$p^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_\gamma(p), \tag{15}$$

where $C_\gamma(p) \in \mathcal{P}(\Omega)$ is described as follows. Each internal node of γ represents a collision and the tree γ describes the collision history. Then $C_\gamma(p)$ represents the distribution obtained at the root after all collisions from γ have been performed, starting with the distribution p at each leaf of γ.

For example, if $k = 3$, there are only two trees $\gamma, \gamma' \in \Gamma(3)$, with $\alpha_3(\gamma) = \alpha_3(\gamma') = 1/2$ and

$$C_\gamma(p) = (p \circ p) \circ p = p \circ (p \circ p) = C_{\gamma'}(p). \tag{16}$$

We refer to Figure 1 for two examples of $C_\gamma(p)$ for $\gamma \in \Gamma(4)$.

Since each collision is of the form $q \circ q'$ for some $q, q' \in \mathcal{P}(\Omega)$ and since by (9) one has

$$p \circ q = \sum_{A \subset [n]} \bar{\nu}(A)\, p_A \otimes q_{A^c}, \tag{17}$$

where $\bar{\nu}(A) = \frac{1}{2}(\nu(A) + \nu(A^c))$, one can write the following expansion for the resulting measure at the root:

$$C_\gamma(p) = \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) C_\gamma^{\vec{A}}(p), \tag{18}$$

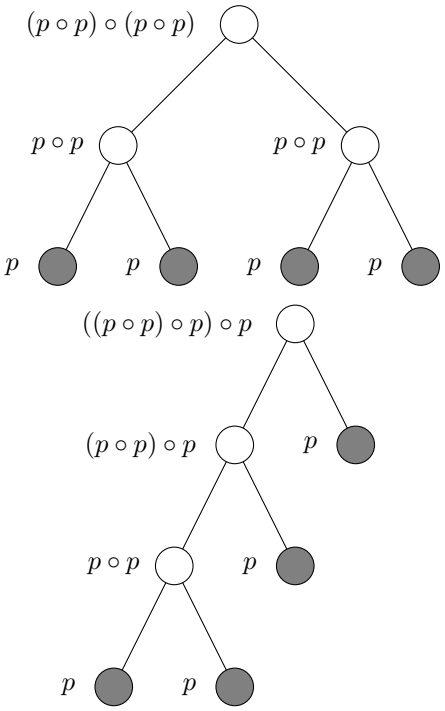


Figure 1: Two possible trees $\gamma, \gamma' \in \Gamma(4)$, and the corresponding distributions $C_\gamma(p) = (p \circ p) \circ (p \circ p)$ and $C_{\gamma'}(p) = ((p \circ p) \circ p) \circ p$.

where $\mathcal{V}_n$ is the set of subsets of $[n]$, so that $\vec{A} \in \mathcal{V}_n^{k-1}$ represents a pattern $\vec{A} = (A_1, \ldots, A_{k-1})$,

$$\nu(\vec{A}) = \prod_{i=1}^{k-1} \bar{\nu}(A_i) \tag{19}$$

and $C_\gamma^{\vec{A}}(p) \in \mathcal{P}(\Omega)$ is the distribution computed as follows. Each internal node v_i, $i = 1, \ldots, k-1$, in γ (in some fixed order) is associated with the set A_i, and we attach the mark A_i to the edge connecting v_i to its left child and the mark A_i^c to the edge connecting v_i to its right child in γ. In this way we obtain a tree γ with marks on all its edges. Let d_i denote the depth of the ith leaf of γ (e.g. counting from the leftmost leaf), and consider $p_{V_i(\vec{A})}$, the marginal of the measure p on the subset

$$V_i(\vec{A}) := \cap_{j=1}^{d_i} A_j^i, \tag{20}$$

where $A_1^i, \ldots, A_{d_i}^i$ are the marks encountered along the edges of the unique path from the root to the ith leaf. Then $C_\gamma^{\vec{A}}(p)$ is given by

$$C_\gamma^{\vec{A}}(p) = p_{V_1(\vec{A})} \otimes \cdots \otimes p_{V_k(\vec{A})}. \tag{21}$$

Note that some of the $V_i(\vec{A})$ may be empty. However, they form a partition of $[n]$, namely $V_i(\vec{A}) \cap V_j(\vec{A}) = \emptyset$ for $i \neq j$ and $\cup_{i=1}^k V_i(\vec{A}) = [n]$, which can be seen as the result of a fragmentation process

$$[n] \to (A_1, A_1^c) \to (A_1 \cap A_2, A_1 \cap A_2^c, A_1^c) \to \cdots \to \Big(V_1(\vec{A}), \cdots, V_k(\vec{A})\Big).$$

As an example, in the case (16) we have $\vec{A} = (A_1, A_2)$, and

$$(p \circ p) \circ p = \sum_{A_1,\, A_2 \subset [n]} \bar{\nu}(A_1)\bar{\nu}(A_2) p_{A_1 \cap A_2} \otimes p_{A_1 \cap A_2^c} \otimes p_{A_1^c}. \tag{22}$$

The proof of (15)-(18) can be easily done by induction, by splitting the tree $\gamma \in \Gamma(k)$ into the left and right subtrees γ_l, γ_r and using the relation (14), see the proof of (48) below for a closely related explicit computation.

In conclusion, from (12), (15) and (18) we obtain the following representation of the distribution p_t as a convex combination of distributions $C_\gamma^{\vec{A}}(p)$:

$$p_t = \sum_{k=1}^{\infty} \beta_t(k) \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A})\, C_\gamma^{\vec{A}}(p), \tag{23}$$

where $\beta_t(k) := e^{-t}(1 - e^{-t})^{k-1}$ is a probability on $\mathbb{N}$ for each $t \geqslant 0$, $\alpha_k(\gamma)$ is a probability on $\Gamma(k)$ for all k, and $\nu(\vec{A})$ is a probability on $\mathcal{V}_n^{k-1}$ for all k.

0.3 Contractive estimates for the nonlinear semigroup

It is convenient to use the notation $S_t(p) = p_t$ for the solution of (11) with initial datum $p_0 = p$, so that $S_{t+s} = S_t S_s = S_s S_t$, for all $s, t \geqslant 0$. This defines the nonlinear semigroup $\{S_t,\, t \geqslant 0\}$. We show a contraction property for S_t when the two initial distributions p, q have the same marginals, that is $p_i = q_i$ for all $i \in [n]$.

Theorem 0.2. *For any probability measure $p, q \in \mathcal{P}(\Omega)$ such that $p_i = q_i$ for all $i \in [n]$, all $t \geqslant 0$,*

$$\|S_t(p) - S_t(q)\|_{\mathrm{TV}} \leqslant \tfrac{1}{2}\, n(n-1)\, \|p - q\|_{\mathrm{TV}}\, e^{-D(\nu)t} \tag{24}$$

where $D(\nu) = 1 - r(\nu)$.

Proof. From (23) we write

$$S_t(p) - S_t(q) = \sum_{k=1}^{\infty} \beta_t(k) s^{(k)}, \tag{25}$$

where

$$s^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \left[C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q) \right]. \tag{26}$$

We introduce some further notation in order to handle more explicitly the difference of probability measures $C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q)$. For any given $\gamma \in \Gamma(k)$, and $\vec{A} \in \mathcal{V}_n^{k-1}$, $\phi : \Omega \mapsto \mathbb{R}$, we may write

$$\begin{aligned} &\langle C_\gamma^{\vec{A}}(p) - C_\gamma^{\vec{A}}(q), \phi \rangle \\ &\quad = \sum_{(x,y) \in \Omega^k \times \Omega^k} \mu(x,y) \sum_{z \in \Omega} \phi(z) \left(\mathbf{1}\left(z = u(y, \vec{A}) \right) - \mathbf{1}\left(z = u(x, \vec{A}) \right) \right) \end{aligned} \tag{27}$$

where $x = (x^i, i = 1, \ldots, k) \in \Omega^k$ stands for the configurations sampled with $q^{\otimes k}$ over the leaves of γ, $y = (y^i, i = 1, \ldots, k) \in \Omega^k$ stands for the configurations sampled with $p^{\otimes k}$ over the leaves of γ, and $\mu(x,y) = \prod_{i=1}^k \mu_1(x^i, y^i)$ denotes a coupling of these two product measures, such that for every $i = 1, \ldots, k$, $\mu_1(x^i, y^i)$ is the optimal coupling of (q, p):

$$\|p - q\|_{\mathrm{TV}} = \sum_{(x,y) \in \Omega^k \times \Omega^k} \mu(x,y) \mathbf{1}\left(x^i \neq y^i \right) = \sum_{x^i, y^i \in \Omega} \mu_1\left(x^i, y^i \right) \mathbf{1}\left(x^i \neq y^i \right). \tag{28}$$

The notation $u(x, \vec{A})$ in (27) is defined as follows. Note that $x^i \in \Omega$ is a vector $x^i = (x_1^i, \ldots, x_n^i)$, for every i, with $x_\ell^i \in \{0, \ldots, q_\ell\}$, and write $x_A^i = (x_\ell^i)_{\ell \in A}$ for the content of x^i on $A \subset [n]$. With this notation, given $x \in \Omega^k$ and $\vec{A} \in \mathcal{V}_n^{k-1}$, $u(x, \vec{A}) \in \Omega$ is defined as the unique configuration such that

$$u(x, \vec{A})_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}$$

for every $i = 1, \ldots, k$. In words, for each $i \in [k]$, the content of the configuration $u(x, \vec{A})$ on the set $V_i(\vec{A}) \subset V$ is taken from the configuration x^i at the ith leaf. The validity of (27) is thus a consequence of (21).

Let $\partial\gamma$ denote the set of leaves of γ. Notice that

$$\mathbf{1}(z = u(x, \vec{A})) = \prod_{i \in \partial\gamma} \mathbf{1}(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})})$$

Let $F = F(\vec{A}, \gamma) \subset \partial\gamma$ denote the set of leaves $i \in \partial\gamma$ such that $|V_i(\vec{A})| > 1$, and write

$$\mathbf{1}(z = u(x, \vec{A})) = w_F(x, z, \vec{A}) w_{F^c}(x, z, \vec{A}),$$

where for any $S \subset \partial\gamma$ we write

$$w_S(x, z, \vec{A}) = \prod_{i \in S} \mathbf{1}(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}). \tag{29}$$

We write $X = (X^i)_{i=1,\dots,k}$ and $Y = (Y^i)_{i=1,\dots,k}$ for the random variables with distribution $q^{\otimes k}$ and $p^{\otimes k}$ respectively. Using the fact that μ is a product over the leaves, and the fact that p, q have the same marginals, for fixed $z \in \Omega$ we have

$$\begin{aligned}\mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] &= \sum_{(x,y)} \mu(x, y) \prod_{i \in F^c} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right) \\ &= \sum_{(x,y)} \mu(x, y) \prod_{i \in F^c} \mathbf{1}\left(z_{V_i(\vec{A})} = y^i_{V_i(\vec{A})}\right) = \mu\left[w_{F^c}(Y, z, \vec{A})\right],\end{aligned} \tag{30}$$

and

$$\begin{aligned}\mu\left[w_{\partial\gamma}\left(Y, z, \vec{A}\right)\right] &= \sum_{(x,y)} \mu(x, y) \mathbf{1}\left(z = u(y, \vec{A})\right) \\ &= \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[w_F\left(Y, z, \vec{A}\right)\right].\end{aligned} \tag{31}$$

With this notation we rewrite (27) as

$$\begin{aligned}&\left\langle C^{\vec{A}}_\gamma(p) - C^{\vec{A}}_\gamma(q), \phi \right\rangle \\ &\quad= \sum_{z \in \Omega} \phi(z) \left[\mu[w_{\partial\gamma}\left(Y, z, \vec{A}\right)] - \mu[w_{\partial\gamma}\left(X, z, \vec{A}\right)\right], \\ &\quad= \sum_{z \in \Omega} \phi(z) \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[w_F\left(Y, z, \vec{A}\right) - w_F\left(X, z, \vec{A}\right)\right].\end{aligned} \tag{32}$$

Note that $w_F(Y, z, \vec{A})] \neq w_F(X, z, \vec{A})$ implies that there exists $i \in F$ such that $X^i \neq Y^i$. Therefore, from (28) we see that

$$\begin{aligned}&\sum_{z \in \Omega} \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \left|\mu\left[w_F\left(Y, z, \vec{A}\right) - w_F\left(X, z, \vec{A}\right)\right]\right| \\ &\leqslant \sum_{i \in F} \sum_{z \in \Omega} \mu\left[w_{F^c}\left(X, z, \vec{A}\right)\right] \mu\left[\left|w_F\left(Y, z, \vec{A}\right) - w_F\left(X, z, \vec{A}\right)\right| \mathbf{1}\left(X^i \neq Y^i\right)\right] \\ &\leqslant 2 \sum_{i \in F} \mu\left[X^i \neq Y^i\right] = 2|F| \, \|p - q\|_{\mathrm{TV}}.\end{aligned} \tag{33}$$

Therefore, we have shown that (26) satisfies

$$\left|\left\langle s^{(k)}, \phi \right\rangle\right| \leqslant 2 \|\phi\|_\infty \|p - q\|_{\mathrm{TV}} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \, |F|. \tag{34}$$

To estimate (34), let us call $\mathcal{A}_j = \mathcal{A}_j(\gamma)$ the set of $\vec{A} \in \mathcal{V}_n^{k-1}$ such that $|V_j(\vec{A})| \geqslant 2$, and let $\mathcal{A} = \cup_{j=1}^k \mathcal{A}_j$. If $\vec{A} \notin \mathcal{A}$ then all sets $V_j(\vec{A})$ are either empty or a single site and therefore $|F| = 0$. Moreover, $|F| = \sum_{j=1}^k \mathbf{1}(\mathcal{A}_j)$. Therefore we write

$$\sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A})\, |F| = \sum_{j=1}^k \nu(\mathcal{A}_j). \tag{35}$$

To estimate the probability $\nu(\mathcal{A}_j)$, observe that the event $\mathcal{A}_j$ implies that there exists $i_1 < i_2 \in [n]$ such that $\{i_1, i_2\} \in V_j(\vec{A})$. Moreover, recalling (20),

$$\nu\left(\{i_1, i_2\} \in V_j(\vec{A})\right) = \nu\left(\{i_1, i_2\} \in \cap_{l=1}^{d_j} A_l^j\right) = \bar{\nu}\left(\{i_1, i_2\} \in A\right)^{d_j} \leqslant \left(\frac{r(\nu)}{2}\right)^{d_j},$$

where we use the fact that for any $i_1 < i_2$, the probability under $\bar{\nu}$ that $\{i_1, i_2\} \in A$ is bounded by $r(\nu)/2$. Indeed, conditionally on not being separated the probability that both i_1, i_2 belong to A under $\bar{\nu}$ is $1/2$ by symmetry. A union bound then shows that

$$\nu(\mathcal{A}_j) \leqslant \tfrac{1}{2}\, n(n-1) \left(\frac{r(\nu)}{2}\right)^{d_j}.$$

In conclusion, if we define

$$\omega(\gamma) := \sum_{j=1}^k \left(\frac{r(\nu)}{2}\right)^{d_j}, \tag{36}$$

we obtain, for all $k \in \mathbb{N}$,

$$\left|\left\langle s^{(k)}, \phi \right\rangle\right| \leqslant n(n-1)\, \|\phi\|_\infty\, \|p - q\|_{\mathrm{TV}} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\, \omega(\gamma). \tag{37}$$

From [7, Lemma 1.4], one has, for all $\varepsilon > 0$,

$$\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\omega(\gamma) \leqslant B_\varepsilon e^{-a_\varepsilon \log k}. \tag{38}$$

with $a_\varepsilon = (1 - r(\nu))/(1 + \varepsilon)$ and $B_\varepsilon = B_\varepsilon(r(\nu), \varepsilon)$. One could use this estimate to obtain an almost optimal decay rate a_ε. However, here we obtain the optimal exponential decay rate by computing the expectation of $\omega(\gamma)$.

Lemma 0.3. *For any $t \geqslant 0$,*

$$\sum_{k=1}^\infty \beta_t(k) \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\, \omega(\gamma) = e^{-(1 - r(\nu))t}. \tag{39}$$

Assuming the validity of (39), and using $\|\mu - \mu'\|_{\mathrm{TV}} = \frac{1}{2} \max_{f:\, \|f\|_\infty \leqslant 1} |\mu(f) - \mu'(f)|$, from (25) and (37) we obtain

$$\|S_t(p) - S_t(q)\|_{\mathrm{TV}} \leqslant \tfrac{1}{2}\, n(n-1)\, \|p - q\|_{\mathrm{TV}}\, e^{-(1 - r(\nu))t}, \tag{40}$$

which concludes the proof of Theorem 0.2.

It remains to prove (39). For any $\gamma \in \cup_{k=1}^{\infty}\Gamma(k)$, define $\mathbb{P}_t(\gamma) = \beta_t(k)\alpha_k(\gamma)$, where k is such that $\gamma \in \Gamma(k)$. Notice that for any $\gamma \in \cup_{k=1}^{\infty}\Gamma(k)$,

$$\mathbb{P}_t(\gamma) = \mathbf{1}_{\gamma=\emptyset}\, e^{-t} + \mathbf{1}_{\gamma\neq\emptyset} \int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)ds, \tag{41}$$

where $\emptyset$ denotes the empty tree with one leaf (given by the root), while for $\gamma \neq \emptyset$ we write γ_l, γ_r for the left and right subtrees. Indeed, for $\gamma = \emptyset$ (41) is immediate, while for $\gamma \neq \emptyset$, supposing $\gamma \in \Gamma(k)$, and recalling (14) one has

$$\begin{aligned}\mathbb{P}_t(\gamma) &= e^{-t}\left(1-e^{-t}\right)^{k-1}\alpha_k(\gamma) = e^{-t}\int_0^t e^{-(t-s)}\left(1-e^{-(t-s)}\right)^{k-2}(k-1)\alpha_k(\gamma)ds \\ &= \int_0^t e^{-s}e^{-2(t-s)}\left(1-e^{-(t-s)}\right)^{k-2}\alpha_j(\gamma_l)\alpha_{k-j}(\gamma_r)ds = \int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)ds.\end{aligned}$$

To prove Lemma 0.3, note that the left hand side of (39) is given by $\sum_\gamma \mathbb{P}_t(\gamma)\omega(\gamma)$. Using (41) and $\omega(\emptyset) = 1$, we see that

$$\begin{aligned}\zeta(t) :&= \sum_\gamma \mathbb{P}_t(\gamma)\omega(\gamma) = e^{-t} + \frac{r(\nu)}{2}\sum_{\gamma\neq\emptyset}\int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma_l)\mathbb{P}_{t-s}(\gamma_r)(\omega(\gamma_l)+\omega(\gamma_r))ds \\ &= e^{-t} + r(\nu)\sum_\gamma \int_0^t e^{-s}\mathbb{P}_{t-s}(\gamma)\omega(\gamma)ds = e^{-t} + r(\nu)\int_0^t e^{-s}\zeta(t-s)ds.\end{aligned}$$

Differentiating, and integrating by parts the resulting expression one finds $\dot\zeta(t) = -(1-r(\nu))\zeta(t)$. Since $\zeta(0) = 1$ we obtain $\zeta(t) = e^{-(1-r(\nu))t}$, $t \geqslant 0$. □

The proof of Theorem 0.1 requires several steps. We start by giving an explicit representation of the solution $\bar S_t(q)(p-q)$ to (6) when $h = p - q$. To this end let us define, for any $k \geqslant 2$, $\gamma \in \Gamma(k)$, the distribution $C_{\gamma,p}(q)$ as the measure defined in (18) with the only difference that we take the distribution p instead of q at the rightmost leaf of γ, all other leafs remaining with the distribution q. Formally,

$$C_{\gamma,p}(q) = \sum_{\vec A \in \mathcal{V}_n^{k-1}} \nu(\vec A)\, C^{\vec A}_{\gamma,p}(q), \qquad C^{\vec A}_{\gamma,p}(q) = q_{V_1(\vec A)} \otimes \cdots \otimes q_{V_{k-1}(\vec A)} \otimes p_{V_k(\vec A)}. \tag{42}$$

In what follows we denote by $d_r(\gamma)$ the depth of the rightmost leaf in γ.

Lemma 0.4. *For all $p, q \in \mathcal{P}(\Omega)$, $t \geqslant 0$,*

$$\bar S_t(q)(p-q) = \sum_{k=1}^{\infty}\beta_t(k)\sum_{\gamma\,\in\,\Gamma(k)}\alpha_k(\gamma)\left(C_{\gamma,p}(q) - C_\gamma(q)\right)2^{d_r(\gamma)}. \tag{43}$$

Proof. Fix $p, q \in \mathcal{P}(\Omega)$. If $h_t = \bar S_t(q)(p)$, then

$$\partial_t \sum_\sigma h_t(\sigma) = 2\sum_\sigma \widehat Q(q_t, h_t)(\sigma) = 0.$$

Thus $\sum_\sigma h_t(\sigma) = 1$, and $2\widehat{Q}(q_t, h_t) = 2q_t \circ h_t - q_t - h_t$. Therefore h_t satisfies

$$h_t = qe^{-t} + \int_0^t e^{-(t-s)} \left(2q_s \circ h_s - q_s\right) ds. \tag{44}$$

By linearity, $\bar{S}_t(q)(p-q) = \bar{S}_t(q)(p) - \bar{S}_t(q)(q)$ satisfies $\bar{S}_t(q)(p-q) = u_t - v_t$, where

$$u_t = pe^{-t} + \int_0^t e^{-(t-s)} 2q_s \circ u_s ds, \qquad v_t = qe^{-t} + \int_0^t e^{-(t-s)} 2q_s \circ v_s ds. \tag{45}$$

Wild's construction then shows that

$$u_t = e^{-t} \sum_{k=1}^{\infty} \left(1 - e^{-t}\right)^{k-1} u^{(k)}, \qquad v_t = e^{-t} \sum_{k=1}^{\infty} \left(1 - e^{-t}\right)^{k-1} v^{(k)}, \tag{46}$$

where $u^{(1)} = p$, $v^{(1)} = q$, and for $k \geqslant 2$,

$$u^{(k)} = \frac{1}{k-1} \sum_{j=1}^{k-1} 2q^{(j)} \circ u^{(k-j)}, \qquad v^{(k)} = \frac{1}{k-1} \sum_{j=1}^{k-1} 2q^{(j)} \circ v^{(k-j)}.$$

respectively, and $q^{(j)}$ is defined by (13) with p replaced by q. To check that this representation of the solution holds, let u_t be defined by (46). Then

$$\begin{aligned}
\int_0^t e^{-(t-s)} 2q_s \circ u_s ds &= \sum_{\ell,k=1}^{\infty} \int_0^t \beta_s(k)\beta_s(\ell) e^{-(t-s)} 2q^{(\ell)} \circ u^{(k)} ds \\
&= \sum_{\ell < k} \int_0^t e^{-2s} \left(1 - e^{-s}\right)^{k-2} e^{-(t-s)} 2q^{(\ell)} \circ u^{(k-\ell)} ds \\
&= \sum_{k \geqslant 2} \int_0^t (k-1) e^{-s} \left(1 - e^{-s}\right)^{k-2} e^{-t} u^{(k)} ds \\
&= \sum_{k \geqslant 2} \left(1 - e^{-t}\right)^{k-1} e^{-t} u^{(k)} = u_t - e^{-t} u^{(1)} = u_t - e^{-t} p.
\end{aligned} \tag{47}$$

Thus u_t solves (45). The same applies to v_t. This proves (46).

Next, we prove that

$$u^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p}(q) 2^{d_r(\gamma)}, \qquad v^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_\gamma(q) 2^{d_r(\gamma)} \tag{48}$$

for all $k \geqslant 2$. Let us prove this by using induction over k. If $k = 2$ then $u^{(2)} = 2q \circ p$ and thus the claimed identity holds. If the formula holds for $j \leqslant k-1$, then using (18) and (14),

$$\begin{aligned}
u^{(k)} &= \frac{1}{k-1} \sum_{j=1}^{k-1} 2q^{(j)} \circ u^{(k-j)} \\
&= \frac{1}{k-1} \sum_{j=1}^{k-1} \sum_{\substack{\gamma_l \in \Gamma(j) \\ \gamma_r \in \Gamma(k-j)}} \alpha_j(\gamma_l)\alpha_{k-j}(\gamma_r) C_{\gamma_l}(q) \circ C_{\gamma_r,p}(q) 2^{d_r(\gamma_r)+1} \\
&= \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p}(q) 2^{d_r(\gamma)}.
\end{aligned}$$

This proves the identity (48) for u, and the same argument proves the one for v. The identities (46) and (48) imply (43). □

Next, for any $k \geqslant 2$, $\gamma \in \Gamma(k)$, $j \in [k]$, we define

$$C_{\gamma,p,j}(q) = \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) C^{\vec{A}}_{\gamma,p,j}(q)\,, \qquad C^{\vec{A}}_{\gamma,p,j}(q) := p_{V_j(\vec{A})} \otimes_{\ell \neq j} q_{V_\ell(\vec{A})}, \quad (49)$$

which denotes the distribution obtained from the tree $\gamma \in \Gamma(k)$ when all leaves are given the distribution q except for the jth leaf which takes the distribution p. When $j = k$, that is j is the rightmost leaf, then we have $C_{\gamma,p,k}(q) = C_{\gamma,p}(q)$.

Lemma 0.5. *For any $k \geqslant 2$, for all p, q:*

$$\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) 2^{d_r} C_{\gamma,p}(q) = \sum_{j=1}^{k} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p,j}(q)\,. \quad (50)$$

Proof. By symmetry, it holds for $k = 2$, since in this case (50) says $2q \circ p = q \circ p + p \circ q$. Assume that (50) holds for all $j \leqslant k-1$. Then

$$\begin{aligned}
&\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) 2^{d_r(\gamma)} C_{\gamma,p}(q) \\
&= \frac{2}{k-1} \sum_{j=1}^{k-1} \sum_{\gamma \in \Gamma(j)} \alpha_j(\gamma) C_\gamma(q) \circ \left(\sum_{\gamma' \in \Gamma(k-j)} \alpha_{k-j}(\gamma') 2^{d_r(\gamma')} C_{\gamma',p}(q) \right) \\
&= \frac{2}{k-1} \sum_{j=1}^{k-1} \sum_{\gamma \in \Gamma(j)} \alpha_j(\gamma) C_\gamma(q) \circ \left(\sum_{i=1}^{k-j} \sum_{\gamma' \in \Gamma(k-j)} \alpha_{k-j}(\gamma') C_{\gamma',p,i}(q) \right) \quad (51) \\
&= 2 \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q),
\end{aligned}$$

where $j(\gamma)$ is the number of leaves in the left subtree of γ, and we have used (14). By symmetry,

$$\sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q) = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=1}^{j(\gamma)} C_{\gamma,p,i}(q)\,. \quad (52)$$

Therefore,

$$\begin{aligned}
&2 \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q) \\
&\qquad = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=1}^{j(\gamma)} C_{\gamma,p,i}(q) + \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{i=j(\gamma)+1}^{k} C_{\gamma,p,i}(q) \\
&\qquad = \sum_{i=1}^{k} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) C_{\gamma,p,i}(q).
\end{aligned} \quad (53)$$

The desired identity follows from (51) and (53). □

We can now turn to the proof Theorem of 0.1.

Proof of (7)**.** From Lemma 0.4 and Lemma 0.5, we write

$$\bar{S}_t(q)(p-q) = \sum_{k=1}^{\infty} \beta_t(k)\, \bar{s}^{(k)}, \tag{54}$$

where

$$\bar{s}^{(k)} = \sum_{j=1}^{k} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \nu(\vec{A}) \left[C^{\vec{A}}_{\gamma,p,j}(q) - C^{\vec{A}}_{\gamma}(q) \right]. \tag{55}$$

Let $\mathcal{A}_j$ denote the set of $\vec{A} \in \mathcal{V}_n^{k-1}$ such that $|V_j(\vec{A})| \geqslant 2$. Since p, q have the same marginals, arguing as in the proof of Theorem 0.2, we obtain, or all $\phi : \Omega \mapsto \mathbb{R}$,

$$\left| \left\langle C^{\vec{A}}_{\gamma,p,j}(q) - C^{\vec{A}}_{\gamma}(q), \phi \right\rangle \right| \leqslant 2\|\phi\|_\infty \|p-q\|_{\mathrm{TV}}\, \mathbf{1}(\mathcal{A}_j)\,. \tag{56}$$

Therefore, as in (37) we obtain

$$\left| \left\langle \bar{s}^{(k)}, \phi \right\rangle \right| \leqslant n(n-1)\, \|\phi\|_\infty\, \|p-q\|_{\mathrm{TV}} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\omega(\gamma). \tag{57}$$

From (54) and Lemma 0.3 we conclude the proof of (7). □

Proof of (8)**.** The proof of (8) requires a bit more work. Let us define

$$r_t := S_t(p) - S_t(q) - \bar{S}_t(q)(p-q).$$

From Lemma 0.4 and Lemma 0.5 we see that

$$r_t = \sum_{k=1}^{\infty} \beta_t(k) r^{(k)}, \tag{58}$$

where

$$r^{(k)} = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \left[C_\gamma(p) - C_\gamma(q) - \sum_{i=1}^{k} \left(C_{\gamma,p,i}(q) - C_\gamma(q) \right) \right], \tag{59}$$

We are going to prove that for any $\phi : \Omega \mapsto \mathbb{R}$,

$$\left\langle r^{(k)}, \phi \right\rangle \leqslant \tfrac{1}{16}\, n^3(n-1)(n-2)\|\phi\|_\infty \|p-q\|^2_{\mathrm{TV}} \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma)\omega(\gamma). \tag{60}$$

By the argument in (40) and Lemma 0.3 this is sufficient to end the proof.

With the notation from the proof of Theorem 0.2 we write

$$\begin{aligned} \left\langle r^{(k)}, \phi \right\rangle = \sum_{\gamma \in \Gamma(k)} \alpha_k(\gamma) \sum_{(x,y) \in \Omega^k \times \Omega^k} \sum_{\vec{A} \in \mathcal{V}_n^{k-1}} \mu(x,y)\nu(\vec{A}) \\ \times \sum_{z \in \Omega} \phi(z) \left(\mathbf{1}\left(z = u(y, \vec{A}) \right) - \mathbf{1}\left(z = u(x, \vec{A}) \right) \right. \\ \left. - \sum_{i=1}^{k} \left(\mathbf{1}\left(z = u(x, y, \vec{A}, i \right) - \mathbf{1}\left(z = u(x, \vec{A}) \right) \right) \right), \end{aligned} \tag{61}$$

where we call $u(x,y,\vec{A},j)$ the unique configuration in Ω such that

$$u(x,y,\vec{A},j)_{V_j(\vec{A})} = y^j_{V_j(\vec{A})}\,, \qquad u(x,y,\vec{A},j)_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\,, \ \text{ for every } \ i \neq j.$$

That is, $u(x,y,\vec{A},j)$ coincides with $u(x,\vec{A})$ except that on $V_j(\vec{A})$ its content (if not empty) is taken from y^j. It follows that

$$\mathbf{1}\left(z = u(x,\vec{A})\right) = \prod_{i\,\in\,\partial\gamma} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right)\,,$$

$$\mathbf{1}\left(z = u\left(x,y,\vec{A},j\right)\right) = \mathbf{1}\left(z_{V_j(\vec{A})} = y^j_{V_j(\vec{A})}\right) \prod_{i\,\in\,\partial\gamma:\ i\neq j} \mathbf{1}\left(z_{V_i(\vec{A})} = x^i_{V_i(\vec{A})}\right)\,,$$

where $\partial\gamma$ denotes the set of leaves of γ.

As in the proof of Theorem 0.2, we let $F = F(\vec{A},\gamma) \subset \partial\gamma$ denote the set of leaves $i \in \partial\gamma$ such that $|V_i(\vec{A})| > 1$, and recall that since μ is a product over the leaves, and p, q have the same marginals one has the identities (30) and (31). With this notation we rewrite (61) as

$$\begin{aligned}\left\langle r^{(k)}, \phi\right\rangle = \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A}\,\in\,\mathcal{V}_n^{k-1}} \nu(\vec{A}) \sum_{z\,\in\,\Omega} \phi(z) \\ \times \Big[\mu\left[w_{\partial\gamma}\left(Y,z,\vec{A}\right)\right] - \mu\left[w_{\partial\gamma}\left(X,z,\vec{A}\right)\right] \\ - \sum_{i=1}^{k}\Big(\mu\left[w_{\partial\gamma}\left(X(i,y),z,\vec{A}\right)\right] - \mu\left[w_{\partial\gamma}\left(X,z,\vec{A}\right)\right]\Big)\Big]\,, \quad (62)\end{aligned}$$

where $X(i,y) \in \Omega^k$ denotes the vector whose i^{th} component is Y^i while all other components are X^j. Moreover, for a given choice of $\gamma, \vec{A}, z$, the square bracket in (62) is also equal to

$$\begin{aligned}\mu\left[w_{\partial\gamma}\left(Y,z,\vec{A}\right)\right] - \mu\left[w_{\partial\gamma}\left(X,z,\vec{A}\right)\right] \\ - \sum_{i=1}^{k}\Big(\mu\left[w_{\partial\gamma}\left(X(i,y),z,\vec{A}\right)\right] - \mu\left[w_{\partial\gamma}\left(X,z,\vec{A}\right)\right]\Big) \\ = \mu\left[w_{F^c}\left(X,z,\vec{A}\right)\right]\ \mu\Big[w_F\left(Y,z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right) \\ - \sum_{i\,\in\,F}\Big(w_F\left(X(i,y),z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right)\Big)\Big]\,. \quad (63)\end{aligned}$$

In particular, if $\gamma, \vec{A}$ are such that $F = F(\gamma,\vec{A}) = \emptyset$, then (63) vanishes. Moreover, (63) vanishes also when F is a single leaf. Indeed, if e.g. $F = \{j\}$, then

$$\begin{aligned}w_F\left(Y,z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right) - \sum_{i\,\in\,F}\Big(w_F\left(X(i,y),z,\vec{A}\right) - w_F\left(X,z,\vec{A}\right)\Big) \\ = w_F\left(Y,z,\vec{A}\right) - w_F\left(X(j,y),z,\vec{A}\right) = 0. \quad (64)\end{aligned}$$

Thus, we may restrict to the case of $|F| > 1$. Therefore,

$$\left|\left\langle r^{(k)}, \phi\right\rangle\right| \leqslant \|\phi\|_\infty \sum_{\gamma\,\in\,\Gamma(k)} \alpha_k(\gamma) \sum_{\vec{A}\,\in\,\mathcal{V}_n^{k-1}} \nu(\vec{A})\mathbf{1}(|F|>1)\,\mu\left[\Gamma\left(\gamma,\vec{A},X,Y\right)\right], \quad (65)$$

where we use the notation

$$\Gamma\left(\gamma,\vec{A},X,Y\right)=\sum_{z\,\in\,\Omega}w_{F^c}\left(X,z,\vec{A}\right)\Big|w_F\left(Y,z,\vec{A}\right)-w_F\left(X,z,\vec{A}\right)$$
$$-\sum_{i\,\in\,F}\left(w_F\left(X(i,y),z,\vec{A}\right)-w_F\left(X,z,\vec{A}\right)\right)\Big|. \quad (66)$$

Clearly, if $X=Y$ the expression within absolute values above vanishes. Moreover, the same applies if (X,Y) is such that $X^i=Y^i$ for all $i\in F$ except one leaf $j\in F$ where $X^j\neq Y^j$. Indeed, in this case $X(i,y)=X$ for all $i\in F$, $i\neq j$, and $X(j,y)=Y$. Given X,Y we write $E=E(X,Y)\subset F$ for the set of leaves $i\in F$ where $X^i\neq Y^i$. Then

$$\Gamma\left(\gamma,\vec{A},X,Y\right)=\sum_{z}w_{E^c\cup F^c}\left(X,z,\vec{A}\right)$$
$$\left|w_E\left(Y,z,\vec{A}\right)-w_E\left(X,z,\vec{A}\right)-\sum_{i\,\in\,E}\left(w_E\left(X(i,y),z,\vec{A}\right)-w_E\left(X,z,\vec{A}\right)\right)\right|$$
$$\leqslant\sum_{z}w_{E^c}\left(X,z,\vec{A}\right)\left|w_E\left(Y,z,\vec{A}\right)-\sum_{i\,\in\,E}w_E\left(X(i,y),z,\vec{A}\right)\right|$$
$$\mathbf{1}(|E|>1)+(|E|-1)\mathbf{1}(|E|>1)$$
$$\leqslant(|E|+1)\mathbf{1}(|E|>1)+(|E|-1)\mathbf{1}(|E|>1)$$
$$=2|E|\mathbf{1}(|E|>1)\leqslant 2|F|\mathbf{1}(|E|>1). \quad (67)$$

Next, we use $|F|\leqslant n/2$, since $\cup_{i\in F}V_i(\vec{A})\subset[n]$ and $|V_i(\vec{A})|\geqslant 2$ for all $i\in F$. Since μ is a product over leaves and on each leaf it satisfies (28), a union bound over the set of pairs in F shows that

$$\mu\left[|E|>1\right]\leqslant\tfrac{1}{2}\,|F|(|F|-1)\|p-q\|_{\mathrm{TV}}^2. \quad (68)$$

Therefore,

$$\left|\left\langle r^{(k)},\phi\right\rangle\right|\leqslant\|\phi\|_\infty\|p-q\|_{\mathrm{TV}}^2\sum_{\gamma\,\in\,\Gamma(k)}\alpha_k(\gamma)\nu\left(|F|^2(|F|-1)\right). \quad (69)$$

Next, observe that $|F|\leqslant n/2$, since $\cup_{i\,\in\,F}V_i(\vec{A})\subset[n]$ and $|V_i(\vec{A})|\geqslant 2$ for all $i\in F$. Thus, if $|F|>1$ one has $|F|^2(|F|-1)\leqslant n^2(n-2)/8$. Summarizing,

$$\left|\left\langle r^{(k)},\phi\right\rangle\right|\leqslant\tfrac{1}{8}\,n^2(n-2)\,\|\phi\|_\infty\,\|p-q\|_{\mathrm{TV}}^2\sum_{\gamma\,\in\,\Gamma(k)}\alpha_k(\gamma)\nu(|F|>1). \quad (70)$$

For a given tree γ, the argument in (37) shows that

$$\nu(|F|>1)\leqslant\nu(|F|\geqslant 1)\leqslant\tfrac{1}{2}\,n(n-1)\sum_{i=1}^{k}\left(\frac{r(\nu)}{2}\right)^{d_i(\gamma)}=\tfrac{1}{2}\,n(n-1)\,\omega(\gamma),$$

where we use the fact that the event $F\geqslant 1$ coincides with $\cup_{i=1}^k\mathcal{A}_i$ from the proof of (37). This ends the proof of (60), which completes the proof of (8). □

References

[1] Basile, G.; Benedetto, D.; Caglioti, E.; Bertini, L. Large deviations for a binary collision model: energy evaporation, Math. Eng., Volume 5 (2023) no. 1, 1, p. 12

[2] Barthe, F.; Cordero-Erausquin, D.; Ledoux, M.; Maurey, B. Correlation and Brascamp–Lieb inequalities for Markov semigroups, Int. Math. Res. Not., Volume 2011 (2011) no. 10, pp. 2177-2216

[3] Bonetto, F.; Geisinger, A.; Loss, M.; Ried, T. Entropy decay for the Kac evolution, Commun. Math. Phys., Volume 363 (2018) no. 3, pp. 847-875

[4] Bobkov, S. G.; Tetali, P. Modified logarithmic Sobolev inequalities in discrete settings, J. Theor. Probab., Volume 19 (2006) no. 2, pp. 289-336

[5] Caputo, P.; Martinelli, F. Asymmetric diffusion and the energy gap above the 111 ground state of the quantum XXZ model, Commun. Math. Phys., Volume 226 (2002) no. 2, pp. 323-375

[6] Caputo, P.; Sinclair, A. Entropy production in nonlinear recombination models, Bernoulli, Volume 24 (2018) no. 4B, pp. 3246-3282

[7] Carlen, E. A.; Carvalho, M. C.; Gabetta, E. Central limit theorem for Maxwellian molecules and truncation of the Wild expansion, Commun. Pure Appl. Math., Volume 53 (2000) no. 3, pp. 370-397

[8] Carlen, E. A.; Carvalho, M. C.; Le Roux, J.; Loss, M.; Villani, C. Entropy and chaos in the Kac model, Kinet. Relat. Models, Volume 3 (2010) no. 1, pp. 85-122

[9] Chaintron, L.-P.; Diez, A. Propagation of chaos: a review of models, methods and applications. I: Models and methods, Kinet. Relat. Models, Volume 15 (2022) no. 6, pp. 895-1015

[10] Cattiaux, P.; Guillin, A.; Malrieu, F. Probabilistic approach for granular media equations in the non-uniformly convex case, Probab. Theory Relat. Fields, Volume 140 (2008) no. 1-2, pp. 19-40

[11] Chung, K. L. A course in probability theory, Academic Press Inc., 2000

[12] Csóka, E.; Harangi, V.; Virág, B. Entropy and expansion, Ann. Inst. Henri Poincaré, Probab. Stat., Volume 56 (2020) no. 4, pp. 2428-2444

[13] Durmus, A.; Eberle, A.; Guillin, A.; Zimmer, R. An elementary approach to uniform in time propagation of chaos, Proc. Am. Math. Soc., Volume 148 (2020) no. 12, pp. 5387-5398

[14] Dolera, E.; Gabetta, E.; Regazzini, E. Reaching the best possible rate of convergence to equilibrium for solutions of Kac's equation via central limit theorem, Ann. Appl. Probab., Volume 19 (2009) no. 1, pp. 186-209

[15] Del Moral, P.; Guionnet, A. On the stability of interacting processes with applications to filtering and genetic algorithms, Ann. Inst. Henri Poincaré, Probab. Stat., Volume 37 (2001) no. 2, pp. 155-194

[16] Del Moral, P.; Miclo, L. Genealogies and increasing propagation of chaos for Feynman–Kac and genetic models, Ann. Appl. Probab., Volume 11 (2001) no. 4, pp. 1166-1198

[17] Diaconis, P.; Shahshahani, M. Generating a random permutation with random transpositions, Z. Wahrscheinlichkeitstheor. Verw. Geb., Volume 57 (1981), pp. 159-179

[18] Diaconis, P.; Saloff-Coste, L. Logarithmic Sobolev inequalities for finite Markov chains, Ann. Appl. Probab., Volume 6 (1996) no. 3, pp. 695-750

[19] Erbar, M.; Fathi, M.; Schlichting, A. Entropic curvature and convergence to equilibrium for mean-field dynamics on discrete spaces, ALEA, Lat. Am. J. Probab. Math. Stat., Volume 17 (2020) no. 1, pp. 445-471

[20] Einav, A. On Villani's conjecture concerning entropy production for the Kac master equation, Kinet. Relat. Models, Volume 4 (2011) no. 2, pp. 479-497

[21] Geiringer, H. On the probability theory of linkage in Mendelian heredity, Ann. Math. Stat., Volume 15 (1944), pp. 25-57

[22] Goel, S. Modified logarithmic Sobolev inequalities for some models of random walk, Stochastic Processes Appl., Volume 114 (2004) no. 1, pp. 51-79

[23] Gao, F.; Quastel, J. Exponential decay of entropy in the random transposition and Bernoulli–Laplace models, Ann. Appl. Probab., Volume 13 (2003) no. 4, pp. 1591-1600

[24] Grünbaum, F. A. Propagation of chaos for the Boltzmann equation, Arch. Ration. Mech. Anal., Volume 42 (1971), pp. 323-345

[25] Hauray, M.; Mischler, S. On Kac's chaos and related problems, J. Funct. Anal., Volume 266 (2014) no. 10, pp. 6055-6157

[26] Kac, M. Foundations of kinetic theory, Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, 1954–1955, vol. III, University of California Press (1956), pp. 171-197

[27] Lacker, D. Hierarchies, entropy, and quantitative propagation of chaos for mean field diffusions, Probab. Math. Phys., Volume 4 (2023) no. 2, pp. 377-432

[28] Lyubich, Y. I. Mathematical structures in population genetics, Biomathematics (Berlin), 22, Springer, 1992 (translated from the Russian by D. Vulis and A. Karpov. Edited by E. Akin)

[29] Martínez, S. A probabilistic analysis of a discrete-time evolution in recombination, Adv. Appl. Math., Volume 91 (2017), pp. 115-136

[30] McKean, H. P. jun. Speed of approach to equilibrium for Kac's caricature of a Maxwellian gas, Arch. Ration. Mech. Anal., Volume 21 (1966), pp. 343-367

[31] McKean, H. P. jun. An exponential formula for solving Boltzmann's equation for a Maxwellian gas, Comb. Theory, Volume 2 (1967), pp. 358-382

[32] Mischler, S.; Mouhot, C. Kac's program in kinetic theory, Invent. Math., Volume 193 (2013) no. 1, pp. 1-147

[33] Mischler, S.; Mouhot, C.; Wennberg, B. A new approach to quantitative propagation of chaos for drift, diffusion and jump processes, Probab. Theory Relat. Fields, Volume 161 (2015) no. 1-2, pp. 1-59

[34] Nagylaki, T. Introduction to theoretical population genetics, Biomathematics (Berlin), 21, Springer, 2013

[35] Petrov, V. V. Sums of independent random variables, Walter de Gruyter, 2022

[36] Platkowski, T.; Illner, R. Discrete velocity models of the Boltzmann equation: a survey on the mathematical aspects of the theory, SIAM Rev., Volume 30 (1988) no. 2, pp. 213-255

[37] Rezakhanlou, F. Propagation of chaos for particle systems associated with discrete Boltzmann equatio, Stochastic Processes Appl., Volume 64 (1996) no. 1, pp. 55-72

[38] Rabani, Y.; Rabinovich, Y.; Sinclair, A. A computational view of population genetics, Random Struct. Algorithms, Volume 12 (1998) no. 4, pp. 313-334

[39] Rabinovich, Y.; Sinclair, A.; Wigderson, A. Quadratic dynamical systems, 33rd annual symposium on Foundations of computer science (FOCS)Proceedings, IEEE (1992), pp. 304-313

[40] Salamat, M.; Baake, M.; Baake, E. The general recombination equation in continuous time and its solution, Discrete Contin. Dyn. Syst., Volume 36 (2016) no. 1, pp. 63-95

[41] Sznitman, A.-S. Topics in propagation of chaos, Calcul des probabilités, Ec. d'Été, Saint-Flour/Fr (Lecture Notes in Mathematics), Volume 1464, Springer (1989), pp. 165-251

[42] Tanaka, H. Probabilistic treatment of the Boltzmann equation of Maxwellian molecules, Z. Wahrscheinlichkeitstheor. Verw. Geb., Volume 46 (1978), pp. 67-105

[43] Villani, C. Cercignani's conjecture is sometimes true and always almost true, Commun. Math. Phys., Volume 234 (2003) no. 3, pp. 455-490

[44] Wild, E. On Boltzmann's equation in the kinetic theory of gases, Proc. Camb. Philos. Soc., Volume 47 (1951) no. 3, pp. 602-609